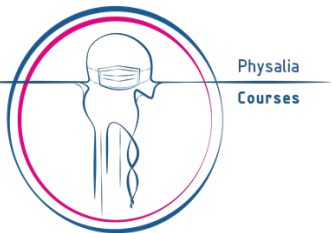
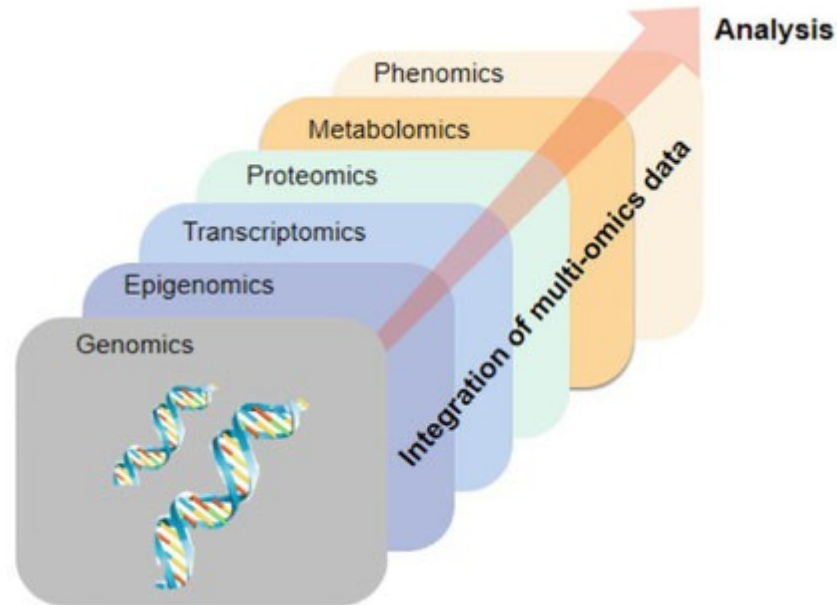


Machine Learning for Multi-Omics Integration

Physalia course

Course outline and practical information

Nikolay Oskolkov, Group Leader (PI) at NIRI, Riga, Latvia



About us

Organizer: Carlo Pecoraro, Physalia courses

info@physalia-courses.org

Instructor:

Dr. Nikolay Oskolkov, MRG Group Leader, NIRI, Riga

nikolay.oskolkov@niri.lv



@NikolayOskolkov



@oskolkov.bsky.social



Personal homepage:
<https://nikolay-oskolkov.com>



Brief introduction: who am I

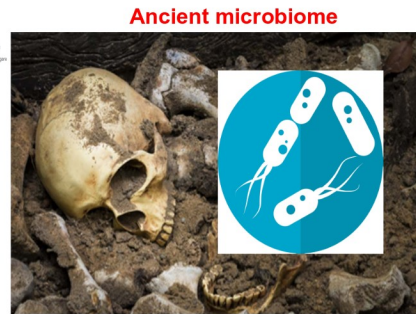
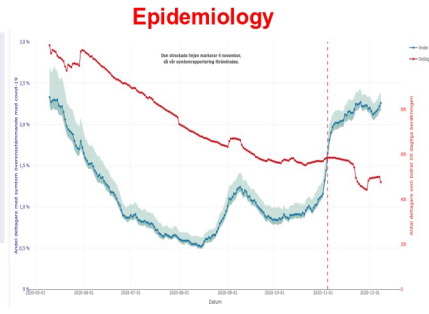
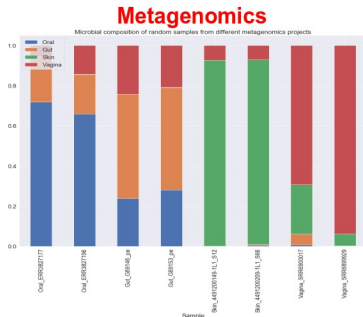
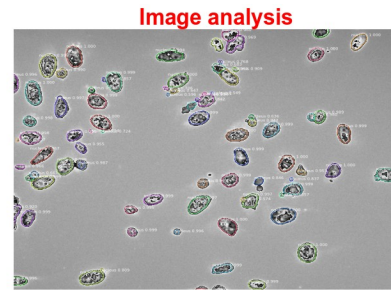
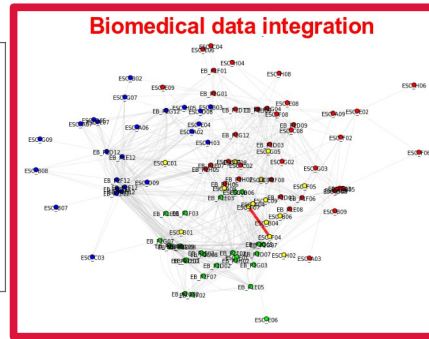
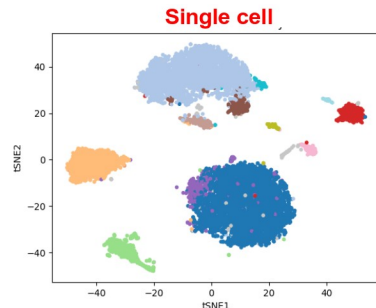
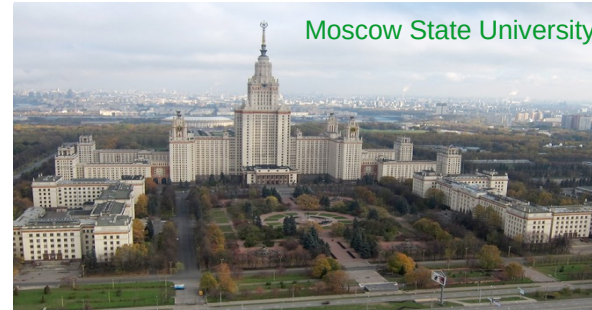
2007 PhD in theoretical physics in Moscow, Russia

2011 medical genomics at Lund University, Sweden

2016 bioinformatician at NBIS SciLifeLab, Sweden

2025 Metabolic Research Group leader, NIRI, Latvia

60+ publications; h-index = 28; 4,600+ citations



NIRI=LIOS+BMC



Metabolic Research Group at NIRI

NIRI NATIONAL
INSTITUTE
OF RESEARCH
AND INNOVATION

Latvian Institute of
Organic Synthesis



NEWS | ABOUT | TEAM | EVENTS | RESEARCH | DISSEMINATION | CONTACT US



Metabolic Research Group

Publications

Conferences

[Metabolic Research Group](#)

The Metabolic Research Group (MRG) focuses on advancing computational methods to identify and validate novel drug targets for metabolic diseases. Our research profile centers on the development and application of machine learning approaches, combined with statistical modeling, to extract biological knowledge from complex datasets. A key expertise of the group is the integration of diverse multiOmics data—including genomics, transcriptomics, proteomics, metabolomics, and metagenomics—enabling a systems-level understanding of metabolic processes and disease mechanisms. Through this integrative and data-driven approach, we aim to contribute to precision medicine by supporting the discovery of innovative therapeutic strategies within the TARGETWISE project.



Ph.D. Nikolay Oskolkov
Group Leader (PI) of the Metabolic
Research Group



Mohammad Banabazi, postdoc
in Metabolic Research Group



Kristina Grausa, PhD student
in Metabolic Research Group

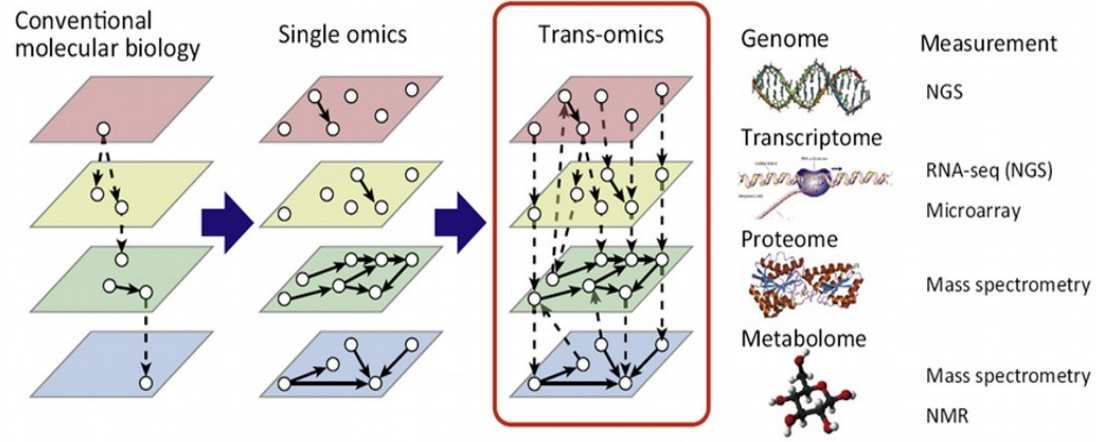


Daniel Rivas, MD, postdoc
in Metabolic Research Group

1 more postdoctoral fellow and 1 PhD student to be hired

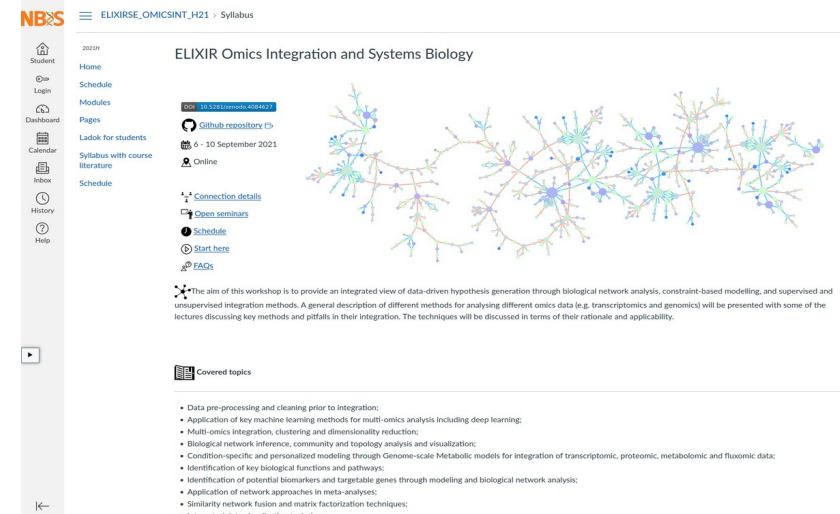
**If you know anyone who might be interested,
please contact me!**

- Name
- University / Institute / Company
- Research interest(s)
- Previous experience with computational analysis and bioinformatics
- Motivation to join the course
- Expectations from the course



scientific reports

Yugi et al., Trends Biotechnol. 2016



https://github.com/NBISweden/workshop_omics_integration

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[nature](#) > [scientific reports](#) > [articles](#) > article

Article | [Open access](#) | Published: 25 June 2024

Predicting type 2 diabetes via machine learning integration of multiple omics from human pancreatic islets

[Tina Rönn](#), [Alexander Perfilyev](#), [Nikolay Oskolkov](#) & [Charlotte Ling](#) ✉

[Scientific Reports](#) 14, Article number: 14637 (2024) | [Cite this article](#)

Rönn et al.,
Scientific Reports 2024

The course was
purchased by



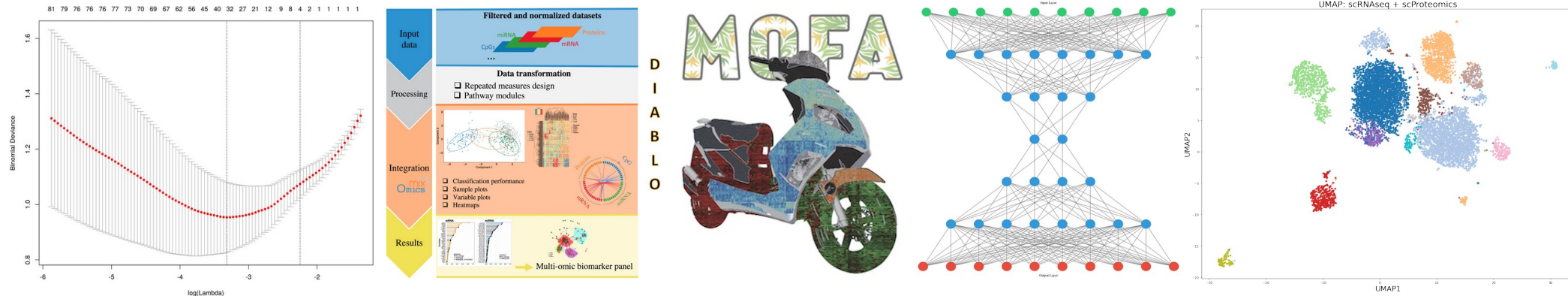
LEADING PROFESSIONAL SOCIETY FOR COMPUTATIONAL
BIOLOGY AND BIOINFORMATICS
CONNECTING, TRAINING, EMPOWERING, WORLDWIDE

Last runs: February 2023 (online), 94 applications
October 2024 (onsite), 49 applications
February 2025 (online), 123 applications

Day 1: introduction of machine learning concepts for data integration, feature selection and supervised Omics integration

Day2: unsupervised Omics integration, factor analysis and graph intersection, neural networks for Omics integration

Day3: dimension reduction methods and UMAP for Omics integration, horizontal and vertical integration of single cell data



PROGRAM

DAY 1: INTRODUCTION & SUPERVISED INTEGRATION

14:00–19:00 (Berlin time)

- 14:00–14:45 | Course overview and introductions
- 15:00–16:00 | Intro to multi-omics ML integration: key concepts
- 16:15–17:15 | Feature selection & supervised Omics integration
- 17:30–18:20 | Feature selection methods: LASSO, PLS, LDA (Lab)
- 18:30–19:00 | Supervised integration using mixOmics and DIABLO (Lab)

DAY 2: UNSUPERVISED & DEEP LEARNING APPROACHES

14:00–19:00 (Berlin time)

- 14:00–15:00 | Unsupervised integration: theory & methods
- 15:15–16:45 | MOFA1 & MOFA2 for unsupervised integration (Lab)
- 17:00–18:00 | Deep Learning for biological data integration
- 18:15–19:00 | Autoencoders for Omics integration (Lab)

DAY 3: SINGLE-CELL OMICS INTEGRATION

14:00–19:00 (Berlin time)

- 14:00–15:00 | UMAP and dimensionality reduction for single-cell data
- 15:15–15:45 | PCA, tSNE, UMAP comparison (Lab)
- 15:45–16:45 | UMAP and graph intersection (Lab)
- 17:00–18:00 | Batch correction & feature integration
- 18:15–19:00 | Seurat CCA + DTW, WNN for single-cell integration (Lab) | Final discussion and Q&A

- Please **ask your questions anytime**, it is ok to interrupt me, or you can raise your hand and ask. You can also **ask questions in the slack workspace** for this course.
- The course includes **6 lectures (~1 h each) and 7 practicals (~1 h each)**, there is a **15-20 min break** after each lecture and each practical. We can potentially make longer breaks if needed.
- During practicals, we will use **Rmarkdown and Jupyter notebooks**. At the end of each practical, I will use rendered html-versions of the notebooks to go through command lines with my explanations.
- The material is based on data and problems from **computational biology**, however the concepts discussed are general and can be applied for other types of data.

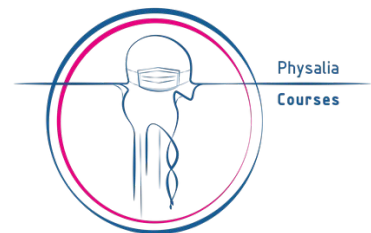
The course will take place from 2 pm to 7 pm (Berlin time)

Links to the Rstudio server will be posted in the Slack channel

The course GitHub repository containing lectures and exercises is:

https://github.com/NikolayOskolkov/Physalia_MLOmicsIntegration_2026

Please bookmark this address!



Physalia_MLOmicsIntegration_2026

Public

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1 Branch

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Q Go to file

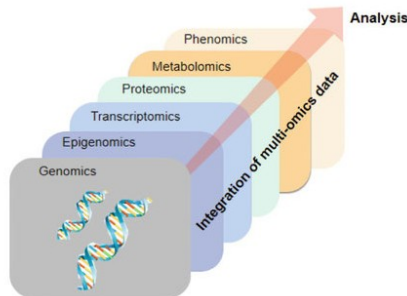
Add file

<> Code

About

NikolayOskolkov	updated README	69d3fc0 · 23 minutes ago	7 Commits
articles	uploaded course material	1 hour ago	
practicals	uploaded course material	1 hour ago	
slides	uploaded course material	1 hour ago	
README.md	updated README	23 minutes ago	
command-line-basics.md	uploaded course material	1 hour ago	
course_logo.jpg	uploaded course material	1 hour ago	
data.zip	uploaded course material	1 hour ago	

README



Machine Learning for Multi-Omics Integration

Instructor

- Dr. Nikolay Oskolkov, PhD, Metabolic Research Group Leader, LIOS, Riga, Latvia

This is the repository containing machine learning multiOmics integration material for the Physalia course

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Languages

- HTML 88.9%
- Jupyter Notebook 11.1%

Suggested workflows

Based on your tech stack

Jekyll using Docker image

Configure

Package a jekyll site using the jekyll/builder Docker image.

SLSA Generic generator

Configure

Generate SLSA3 provenance for your existing release workflows

[More workflows](#) [Dismiss suggestions](#)

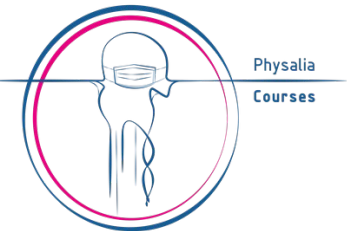
Practical information: Amazon Cloud (AWS EC2)



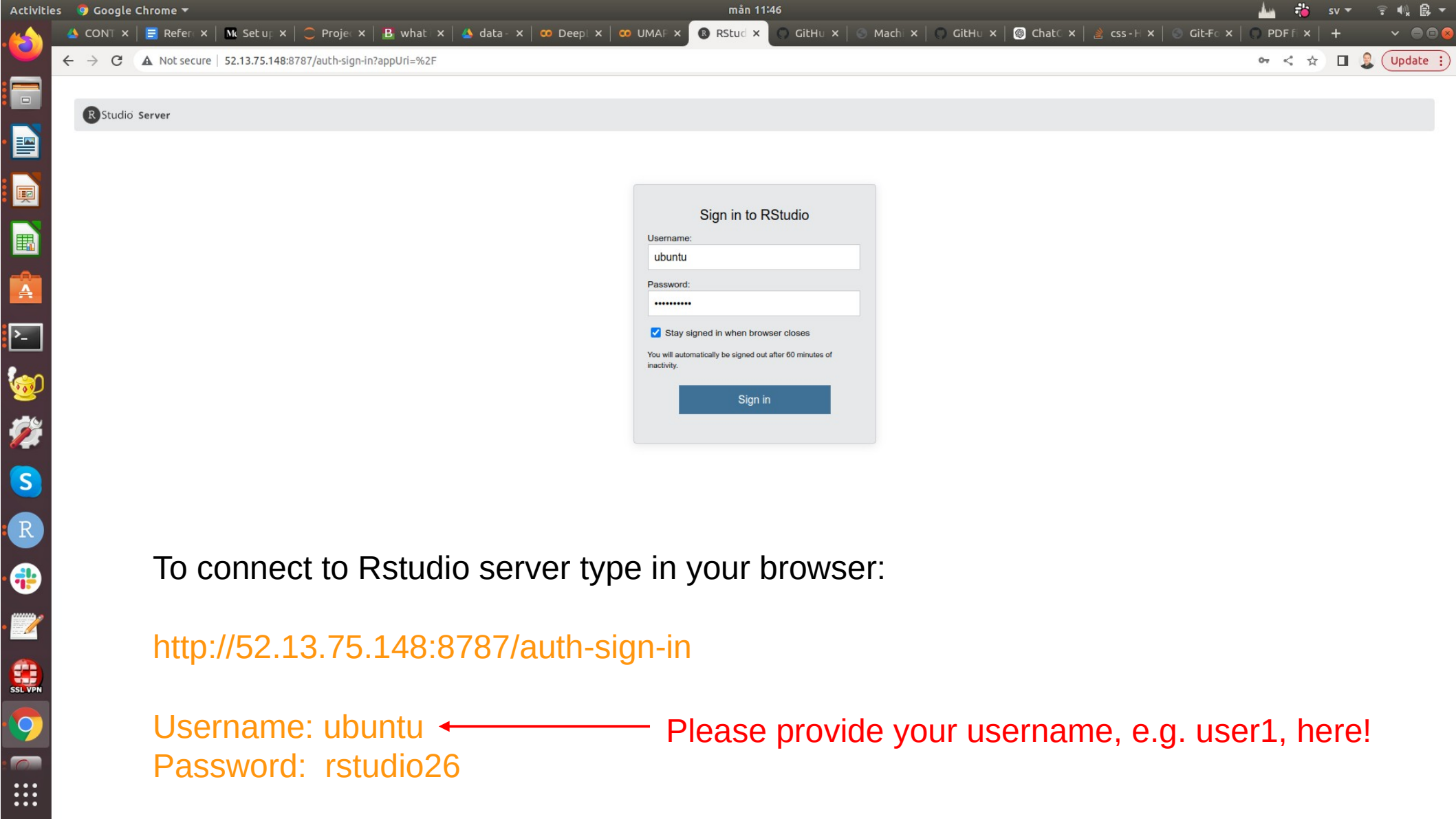
We will use the Cloud Computing service from Amazon, which we will access via `ssh` (secure shell protocol)

```
ssh -i multiomics.pem -X ubuntu@54.244.40.220
```

- The IP address changes every day
- Everyone is given a username, with a `home` and `shared` folders
 - List of usernames can be found in Slack
 - The `shared` folder is copy-only: do not delete, move, rename, or write



However, most of the time you will be using Rstudio server
(for R exercises) and Google Colab (for Python exercises)!



To connect to Rstudio server type in your browser:

<http://52.13.75.148:8787/auth-sign-in>

Username: ubuntu ← Please provide your username, e.g. user1, here!
Password: rstudio26

Activities

Google Chrome

ons 14:33

CONTAMINATION_PAPER x Reference_Contaminatio x r -Launch rstudio-server / x RStudio Server x Install | mixOmics x +

Not secure | 34.217.124.65:8787

ubuntu Project: (None)

R

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Console Terminal Background Jobs

R 4.3.3 ~/
R version 4.3.3 (2024-02-29) -- "Angel Food Cake"
Copyright (C) 2024 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> |

Environment History Connections Tutorial

Import Dataset 137 MiB

R Global Environment

Environment is empty

Files Plots Packages Help Viewer Presentation

New Folder New File Upload Delete Rename More

Home

	Name	Size	Modified
☐	libssl1.1_1.1.1f-1ubuntu2_amd64.deb	1.3 MB	Apr 21, 2020, 4:33 PM
☐	miniconda3		
☐	Miniconda3-latest-Linux-x86_64.sh	140.9 MB	Jan 8, 2025, 12:08 PM
☐	R		
☐	rstudio-server-2024.12.0-467-amd64.deb	122.4 MB	Dec 16, 2024, 8:50 PM
☐	rstudio-server-2024.12.0-467-amd64.deb.1	122.4 MB	Dec 16, 2024, 8:50 PM
☐	snap		

Plan B (in case of network problems): install locally

```
if (!require("BiocManager", quietly = TRUE))  
  install.packages("BiocManager")
```

```
BiocManager::install("mixOmics")  
BiocManager::install("MOFA2")  
install.packages("rmarkdown")  
install.packages("knitr")  
install.packages("glmnet")  
install.packages("Rtsne")  
install.packages("uwot")
```

```
BiocManager::install(c('AnnotationFilter', 'BiocGenerics', 'GenomeInfoDb',  
  'GenomicFeatures', 'GenomicRanges', 'IRanges', 'Rsamtools', 'S4Vectors', 'TFBSTools',  
  'ggbio', 'motifmatchr', 'AnnotationDbi'))
```

```
install.packages("Seurat", dependencies = TRUE)  
install.packages("Signac", dependencies = TRUE)
```

To run Jupyter notebooks on Google Colab please do the following:

1. Clone this repo: https://github.com/NikolayOskolkov/Physalia_MLOmicsIntegration_2026
2. Unzip the data.zip folder locally and upload it to your Google drive
3. Start the Jupyter notebook from the Google drive and mount your google drive:

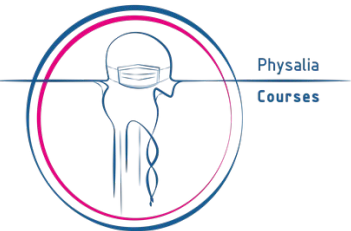
#Mount Google Drive to Colab

```
from google.colab import drive  
drive.mount('/content/drive')
```

```
!ls -l /content/drive/MyDrive/
```

```
import os  
os.chdir("/content/drive/MyDrive/data")
```

N.B. you need a Google account
to be able to run Google Colab!



Google Chrome

CONT x Refer x Mk Set up x Proje x B what x UMAF x DeepL x UMAF x R RStu x GitHu x Mach x GitHu x ChatC x css -F x Git-Fc x PDF f x +

colab.research.google.com/drive/1Q57pB5IT92sRKQEuGERQSGdMVHmC6qZd#scrollTo=HLDfYRzqMzAe

Update

DeepLearningDataIntegration.ipynb

File Edit View Insert Runtime Tools Help All changes saved

+ Code + Text

RAM Disk Gemini

Deep Learning for Data Integration

Biological and biomedical research has been tremendously benefiting last decade from the technological progress delivering DNA sequence (genomics), gene expression (transcriptomics), protein abundance (proteomics) and many other levels of biological information commonly referred to as OMICs. Despite individual OMICs layers are capable of answering many important biological questions, their combination and consequent synergistic effects from their complementarity promise new insights into behavior of biological systems such as cells, tissues and organisms. Therefore OMICs integration represents the contemporary challenge in Biology and Biomedicine.

[] from google.colab import drive
drive.mount('/content/drive')

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

[] !ls -l /content/drive/My Drive/data/DeepLearningDataIntegration

total 49588
-rw-rw-r-- 1 root root 5105 Jan 10 12:26 CITEseq.py
-rw-rw-r-- 1 root root 2066933 Jan 10 12:26 DeepLearningDataIntegration.html
-rw-rw-r-- 1 root root 1747228 Jan 10 12:41 DeepLearningDataIntegration.ipynb
-rw-rw-r-- 1 root root 130233 Oct 3 2020 DeepLearningDataIntegration.jpg
-rw-rw-r-- 1 root root 5598515 Jan 10 12:26 DeepLearning_SingleCell_10X_1.3Mcells.html
-rw-rw-r-- 1 root root 3499 Jan 10 12:26 dim_reduct_CITEseq.py
-rw-rw-r-- 1 root root 3616 Jan 10 12:26 dim_reduct_scNMTseq.py
-rw-rw-r-- 1 root root 134229 Aug 8 2019 IntegrOmicsMethods.png
-rw-rw-r-- 1 root root 84 Jan 10 12:26 README.md
-rw-rw-r-- 1 root root 3772209 Jan 10 12:26 scATACseq.txt
-rw-rw-r-- 1 root root 4082712 Jan 10 12:26 scBSseq.txt
-rw-rw-r-- 1 root root 6141 Jan 10 12:26 scNMTseq.py
-rw-rw-r-- 1 root root 467570 Jan 10 12:26 scProteomics_CITEseq.txt
-rw-rw-r-- 1 root root 17128751 Jan 10 12:26 scRNAseq_CITEseq.txt
-rw-rw-r-- 1 root root 15547715 Jan 10 12:26 scRNAseq.txt
-rw-rw-r-- 1 root root 71008 Aug 8 2019 Strategies-for-multi-omics-profiling-of-single-cells-Three-major-types-of-molecules.png
-rw-rw-r-- 1 root root 3647 Jan 10 12:26 tSNE_on_Autoencoder_CITEseq.py
-rw-rw-r-- 1 root root 3635 Jan 10 12:26 tsne_on_autoencoder_scNMTseq.py

import os
os.chdir("/content/drive/My Drive/data/DeepLearningDataIntegration")

[] from IPython.display import Image
Image('DeepLearningDataIntegration.png', width=2000)

Connected to Python 3 Google Compute Engine backend